Student Handout

Name: _____ **Date:** _____ **Class:** _____

MyDesign Portfolio

Element D Initial Template

Guidance before filling out the template on the next page:

Because you will iterate on your design multiple times, you will need to create multiple copies of these sections. Think of Element D as a diary that you will date and keep adding to throughout your work until your final iteration. Iteration must shine through in this Element.

You should begin with many design ideas that reflect the individual ideas of your group members and then the group ideas. Number these ideas and include a date, description, drawing, and details for each.

After you have created an initial set of ideas, document your decision making on which design to include, ideally with a decision matrix.

Create a plan of work, ideally using a Gantt chart that may be housed in this document or in a different day-to-day document.

Generate a design blueprint with enough details that someone else not familiar with your project could also build your design. Think of this as your first sketch.

You will then proceed forward to build and test this design, returning to Element D afterwards when you are ready to create more design ideas. Fill out each of these steps again.

Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation *(D1)*

This section focuses on the process of generating solutions. The initial rounds should include every idea you come up with but the subsequent rounds of ideation should lead to viable options. Each idea should include not only words, but also a picture. Use as many copies of this sub-template as needed.

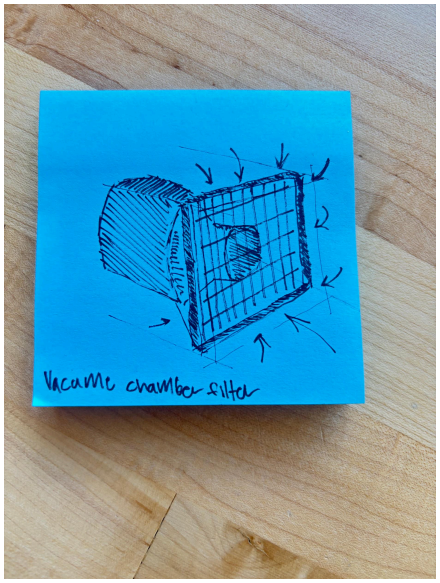
Name: _____ **Date:** _____ **Class:** _____

Design Idea #1

Date: March 4th, 2025
Description: A fine mesh net that will accommodate spacing in between each netting for microplastic collection. We plan to place it either in the middle or on the sides of the ROV.
Drawing that includes distinguishing information such as scale, dimensions, and materials:

Design Idea #2

Date: March 4th, 2025
Description: Vacuum filtration system that captures water and uses suction technology to filter out microplastics while releasing water into the medium.
Drawing that includes distinguishing information such as scale, dimensions, and materials:



Name: _____ **Date:** _____ **Class:** _____

Design Idea Comparison and Selection *(D1, D2)*

When you've finished generating ideas, you must compare your design options according to all your design requirements. Incorporate the priority of those requirements when comparing design options. A decision matrix (perhaps a Pugh Matrix) is encouraged since it allows for systematic, weighted comparisons. If using weights in a decision matrix, be sure to explain and justify lothe weighting based on your design requirements. You may use the weighted design matrix template below, inserting your criteria and design options.

Design Requirement #1:
ROV should be capable of movement in directions of a 360-degree radius.

Design Requirement #2:
ROV should be capable of safely and efficiently collecting microplastics of multiple sizes.

Design Requirement #3:

ROV should be able to reach a max depth of 5 meters. It should be able to control its depth.

Design Requirement #4:

ROV should be capable of buoyancy and resurfacing after emerging underneath the site of interest

Design Requirement #5:

All wiring throughout the system should be waterproofed and tested

	weight	Option 1	Option 2	Option 3
Criterion A	20%	2	2	3
Criterion B	30%	5	4.5	5
Criterion C	15%	3	3	4

Name: _____ **Date:** _____ **Class:** _____

	weight	Option 1	Option 2	Option 3
Criterion D	15%	4	2	3
Criterion E	20%	5	5	4
score		19	16.5	19

All options scored out of 1-5 Scale.

Option 1:A fine mesh net must accommodate spacing between nets for microplastics.
Option 2:A pressurized vacuum tube that would suck microplastics out of water samples.
Option 3: A hydrophobic interaction tube that would automate the separation process of water and microplastics.

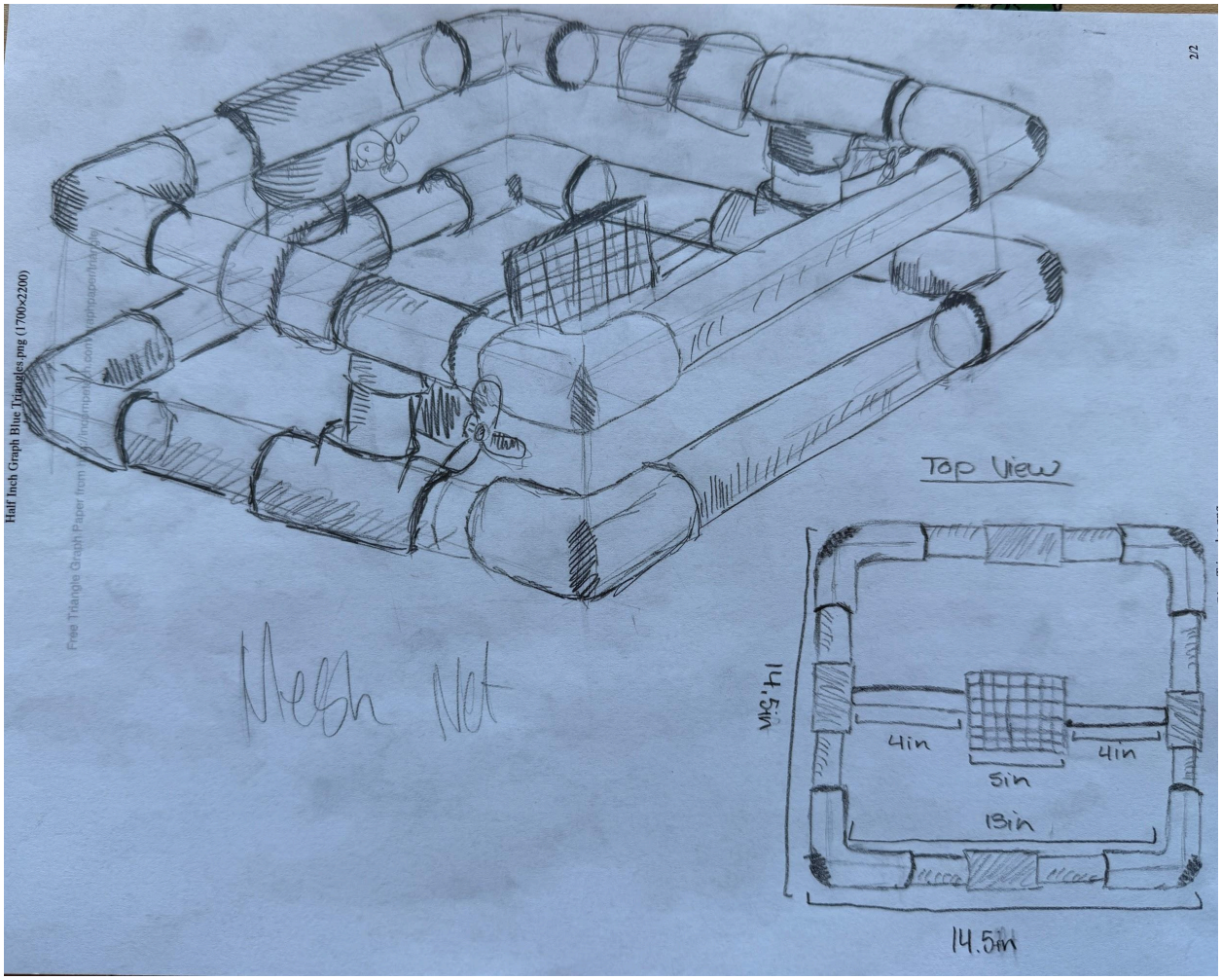
Design Blueprint *(D4)*

Describe all of your design's details such that someone unfamiliar with your project could build your prototype. Include sketches, dimensions, CAD (if taught in your class), manufacturing methods, material choices, supply lists, and costs.

Design Blueprint #1

Date: March 4th, 2025
Sketch:

Name: _____ **Date:** _____ **Class:** _____



Manufacturing Methods:

Step 1: Find a mesh net that accommodates the collection of microplastics.

Step 2: Build pipes out that will hold the microplastic capturing fine mesh net.

Step 3: Figure out an efficient way to hold the net in place by connecting it to the already-built pipes.

Supply List and Costs:

Item	**Cost each**	**# needed**	**Total**
Fine Mesh Net	$10/sqft.	2-3	Shouldn't need more than $10-$15.

Name: _____ **Date:** _____ **Class:** _____

Plan of Action *(D3)*

A Gantt chart is encouraged to lay out the primary steps expected on a visual timeline. Larger tasks from the Gantt chart should be broken down further into detailed sub-tasks, which may be housed in a different day-to-day document. Discrete tasks should include enough detail such that others could replicate the efforts, such as procedures, materials, tools, status, etc.

Ethan, Eero, & Arjun - Underwater ROV Timeline